

SCALE-INVARIANT ROBUST FENCHEL TROPICAL ATTENTION FOR OUT-OF-DISTRIBUTION ALGORITHMIC EXECUTION

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ABSTRACT

Standard self-attention relies on the smooth scaled dot-product softmax kernel, which produces continuous quadratic decision boundaries and suffers from severe attention dispersion under out-of-distribution (OOD) value shifts and input noise. Tropical attention introduces a polyhedral inductive bias by replacing exponentiation with max-plus algebra, yet existing max-plus models collapse under extreme dynamic-range shifts ($X \rightarrow \alpha X$) and noise because of unclamped valuation maps and static devaluation. We introduce **Scale-Invariant Robust Fenchel Tropical Attention (TKFS-Attention)**, a streamlined max-plus attention that unifies five primitives: (i) per-sequence robust Chebyshev scale normalization with outlier clamping, (ii) dual softplus Fenchel projection into the tropical projective simplex, (iii) dual-toric tropical Hilbert projective metric scoring, (iv) exact hard max-plus context aggregation, and (v) gated devaluation. Across all 11 tasks of a CLRS-style algorithmic-execution suite under strict OOD stress protocols, TKFS-Attention attains a mean Micro-F1 of **77.09%** under Value OOD (vs. 71.71% original Tropical, 75.26% Adaptive Softmax, 74.19% Vanilla Softmax) and **75.39%** under Perturbative Noise OOD (vs. 70.98%/69.99%/68.62%). On Convex Hull under Value OOD it improves accuracy from 77.11% (Tropical) and 68.95% (Vanilla) to **97.05%**. Extensive ablations isolate robust median-quantile normalization with clamping and dual softplus valuation as the primary drivers of OOD robustness, while exact hard max-plus routing preserves exact combinatorial geometry.

1 INTRODUCTION

Executing algorithmic reasoning procedures—shortest-path computation, dynamic-programming recurrences, set partitioning, and convex-hull identification—remains a fundamental bottleneck for modern neural architectures (Bengio et al., 2018; Xu et al., 2019; Velivcković et al., 2022). Although Transformers powered by scaled dot-product self-attention dominate language and vision, their capacity for systematic *out-of-distribution* (OOD) algorithmic execution is severely constrained (Zhou et al., 2023; Dziri et al., 2023). The root cause is a mathematical misalignment between the smooth, exponential dot-product kernel and the discrete, piecewise-linear (PWL) geometry intrinsic to combinatorial problems (Montúfar et al., 2014; Zhang et al., 2018). Softmax normalizes query–key interactions via exponentials, inducing smooth quadratic decision boundaries and suffering “attention fading” when exposed to longer sequences, unseen coordinate magnitudes, or input noise (Press et al., 2021; Xiao et al., 2023).

To ground networks in discrete computation, prior work explored graph neural networks (Velicković et al., 2019; Dudzik & Velickovic, 2022), pointer mechanisms (Vinyals et al., 2015; Velivcković et al., 2020), generalist execution engines (Ibarz et al., 2022; Bounsi et al., 2024), and program-execution modules (Reed & de Freitas, 2015; Rodionov & Prokhorenkova, 2023). Recently, *Tropical Attention* (Hashemi et al., 2025) lifted self-attention into tropical projective space: operating natively in the max-plus semiring $(\mathbb{R} \cup \{-\infty\}, \max, +)$, it scores tokens with the tropical Hilbert projective metric and aggregates states via max-plus matrix–vector products. Because every

max-plus operation is piecewise-linear, tropical attention equips Transformers with a *polyhedral inductive bias*, forming sharp decision hyperplanes that mirror the polyhedral complexes of dynamic programming and combinatorial optimization (Zhang et al., 2018; Brandenburg et al., 2024).

Despite this promise, existing tropical attention degrades sharply under extreme Value-OOD magnitude shifts ($X \rightarrow \alpha X$) and adversarial noise. First, prior valuation maps lack scale-invariant normalization: as input magnitudes scale OOD, unclamped valuations inflate variance and distort projective representations. Second, static exponential devaluation amplifies signal noise, hurting continuous targets and noise-corrupted inputs. Third, single-rank tropical routing struggles to capture multi-component graph closures without excessive depth or width.

We address these vulnerabilities with **Scale-Invariant Robust Fenchel Tropical Attention (TKFS-Attention)**, whose contributions are:

1. **Scale-Invariant Robust Fenchel Valuation.** A per-sequence robust Chebyshev scale estimator μ with outlier clamping followed by a dual softplus Fenchel projection $Z_i = [\log(\text{Softplus}(\bar{X}_i)) \parallel \log(\text{Softplus}(-\bar{X}_i))] \in \mathbb{R}^{2d}$, guaranteeing scale invariance under $X \rightarrow \alpha X$ (Lemma 1) while smoothing the log-domain near zero.
2. **Dual-toric Hilbert scoring with exact hard max-plus aggregation**, preserving exact polyhedral decision facets needed for discrete tasks (Convex Hull, Floyd–Warshall).
3. **Gated devaluation**, a learnable per-channel gate g that selects scale-covariant reconstruction under value shifts or scale-free reconstruction under noise, resolving the classification-vs-regression tension.
4. **A comprehensive study** on all 11 CLRS-style tasks under In-Distribution, Value OOD, and Perturbative Noise OOD, plus five ablations dissecting every primitive.

TKFS-Attention improves mean Value-OOD Micro-F1 to **77.09%** and Noise-OOD to **75.39%** over Vanilla (74.19/68.62), Adaptive (75.26/69.99), and original Tropical (71.71/70.98) attention, with the largest gains on scale-sensitive geometry (Convex Hull Value OOD 68.95 $\rightarrow$ 97.05%) and shortest paths (Floyd–Warshall Value OOD 33.88 $\rightarrow$ 57.98%).

2 RELATED WORK

Softmax and adaptive attention for algorithmic execution. The scaled dot-product kernel $\text{Softmax}(QK^\top/\sqrt{d_k})V$ excels at language but fails on combinatorial execution (Zhou et al., 2023; Dziri et al., 2023). Zhou et al. (2023) tie length generalization to whether a task admits a short RASP program, and Dziri et al. (2023) show autoregressive Transformers solve compositional tasks by memorizing subgraph patterns rather than executing algorithms. A key failure mode is exponential *attention dispersion*: as length grows, the softmax denominator sums over more tokens, diluting weights and blurring decisions (Press et al., 2021; Xiao et al., 2023). Adaptive-temperature scaling (Balanya et al., 2022) and in-context function fitting (Garg et al., 2022) mitigate but do not remove this fragility under magnitude shift. TKFS-Attention instead replaces the exponential kernel with a scale-invariant tropical projective kernel, eliminating fading and keeping sharp polyhedral boundaries.

Recurrent Transformers and dynamic halting. To emulate iterative control flow, Adaptive Computation Time (Graves, 2016), Universal Transformers (Dehghani et al., 2018), PonderNet (Banino et al., 2021), and depth-adaptive Transformers (Elbayad et al., 2019) add depth-wise recurrence. These incur large inference latency and unnormalized hidden-state growth under value shift (Trask et al., 2018; Ontan’on et al., 2021). Recent theory frames DP training as distributionally robust control (Akman et al., 2026), uses latent bottlenecks for OOD compositionality (Altabaa et al., 2025), and shows high-confidence attention acts as a tropical matrix product executing shortest-path updates (Alpay & Senturk, 2026). Unlike deep recurrent models (3 $\times$ –9 $\times$ slower), TKFS-Attention absorbs transitive closures into a non-recurrent polyhedral representation.

Tropical neural networks. Tropical geometry over the max-plus semiring gives a rigorous language for PWL networks (Zhang et al., 2018; Maragos et al., 2021; Brandenburg et al., 2024): Zhang et al. (2018) prove ReLU networks are tropical rational maps whose linear regions are tropical-polytope vertices. Max-plus architectures span min-max-plus universal approximators (Luo & Fan, 2021), morphological networks (Mondal et al., 2019), tropical convolutions (Li et al., 2025), adversarially

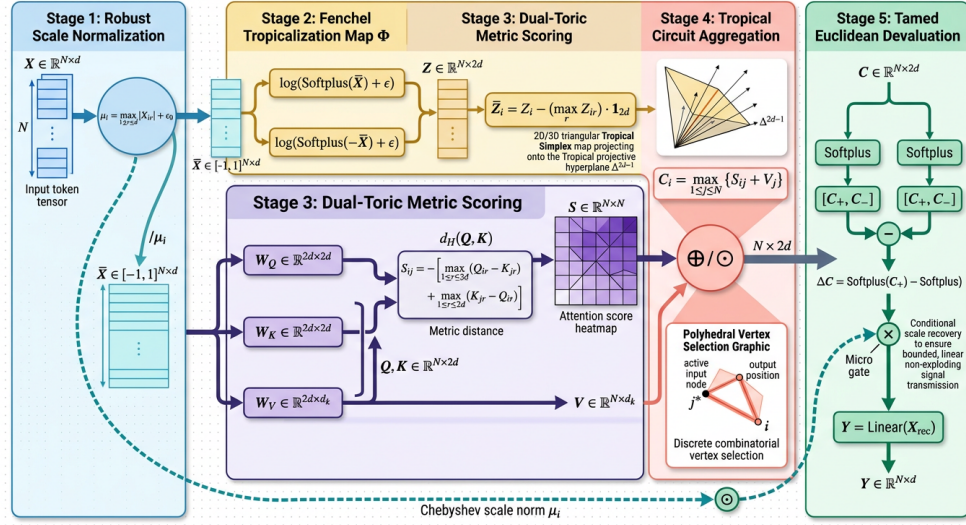


Figure 1: End-to-end pipeline of streamlined TKFS-Attention. **Stage 1** performs robust per-token Chebyshev scale normalization for input scale invariance. **Stage 2** applies a dual softplus Fenchel valuation, projecting features into the tropical projective simplex Δ^{2d-1} . **Stage 3** builds queries/keys/values and computes pairwise non-expansive tropical Hilbert metric scores S_{ij} . **Stage 4** aggregates values by exact hard max-plus routing (discrete combinatorial vertex selection). **Stage 5** maps context back to Euclidean space via a *gated* softplus dual-difference devaluation, conditionally rescaled by the preserved Chebyshev norm to guarantee non-exploding signal transmission under distribution shift.

robust tropical embeddings (Pasque et al., 2024), and tree-space tropical networks (Yoshida et al., 2023). Hashemi et al. (2025) introduced Multi-Head Tropical Attention (MHTA) but its unclamped valuation and unconditional devaluation collapse under $X \rightarrow \alpha X$ and noise. TKFS-Attention directly repairs these with robust Fenchel valuation and gated devaluation.

3 METHODOLOGY

3.1 TROPICAL PROJECTIVE GEOMETRY

Combinatorial algorithms—dynamic programming, greedy selection, convex-hull construction—live naturally in the *tropical semiring* $(\mathbb{T}, \oplus, \otimes)$ with $\mathbb{T} := \mathbb{R} \cup \{-\infty\}$, $a \oplus b := \max(a, b)$, and $a \otimes b := a + b$; the additive/multiplicative identities are $-\infty$ and 0. Matrix–vector products then realize Bellman updates: $(A \odot x)_i = \max_j (A_{ij} + x_j)$. To secure scale invariance, tokens are analyzed in the *tropical projective simplex*, the quotient $\mathbb{R}^d / \mathbb{R}\mathbf{1}_d$ under shifts $z \sim z + c\mathbf{1}_d$:

$$\Delta^{d-1} := \left\{ z \in \mathbb{R}^d \mid \max_{1 \leq i \leq d} z_i = 0 \right\}, \quad \pi(x) = x - \left(\max_i x_i \right) \mathbf{1}_d. \quad (1)$$

Because every tropical operation is a PWL composition of max and +, decision boundaries are exact polyhedral complexes of linear hyperplanes (Zhang et al., 2018; Brandenburg et al., 2024), sharply aligned with combinatorial facets—unlike smooth softmax boundaries that disperse across tokens. Figure 1 shows the five-stage TKFS pipeline.

3.2 SCALE-INVARIANT ROBUST FENCHEL VALUATION

Standard tropical valuations $\log(\max(0, X))$ fail under $X \rightarrow \alpha X$ and have zero-gradient singularities for non-positive inputs. Given $X \in \mathbb{R}^{N \times d}$, we first estimate a robust per-sequence scale via a quantile of the per-token Chebyshev norms,

$$\mu_0 = \text{Quantile}_q \left(\left\{ \max_{1 \leq r \leq d} |X_{ir}| \right\}_{i=1}^N \right), \quad \mu = \mu_0 + \epsilon_0, \quad (2)$$

with $q=0.5$ (sequence median, robust to outlier/noise spikes) and floor $\epsilon_0=10^{-4}$; μ_0 is scale-homogeneous, $\mu_0(\alpha X) = \alpha \mu_0(X)$. Tokens are normalized with outlier clamping ($c=1.0$) and

mapped into a dual $2d$ -space by a bi-conjugate Fenchel softplus decomposition, then projected onto the simplex:

$$\bar{X}_i = \text{clamp}(X_i/\mu, -c, c) \in [-c, c]^d, \quad Z_i = [\log(\text{Softplus}(\bar{X}_i) + \epsilon) \parallel \log(\text{Softplus}(-\bar{X}_i) + \epsilon)], \quad (3)$$

$$\bar{Z}_i = Z_i - (\max_{1 \leq r \leq 2d} Z_{ir}) \mathbf{1}_{2d} \in \Delta^{2d-1}, \quad \epsilon = 10^{-6}. \quad (4)$$

The dual softplus handles negative components while remaining differentiable through zero.

Lemma 1 (Projective scale invariance). *As $\epsilon_0 \rightarrow 0$, for any $\alpha > 0$ and non-zero X , $\bar{Z}(\alpha X) = \bar{Z}(X)$. For finite ϵ_0 , the drift is bounded by $\mathcal{O}(\epsilon_0/(\alpha\mu_0(X) + \epsilon_0))$.*

Proof. See Appendix A. Empirically the deployed drift under a $10\times$ shift is ≤ 0.041 (Table 7).

3.3 DUAL-TORIC METRIC SCORING AND HARD MAX-PLUS AGGREGATION

From $\bar{Z} \in \mathbb{R}^{N \times 2d}$, projections $W_Q, W_K \in \mathbb{R}^{2d \times 2d}$, $W_V \in \mathbb{R}^{2d \times d_k}$ give Q, K, V . Compatibilities use the *dual-toric tropical Hilbert projective metric*:

$$S_{ij} = -d_H(Q_i, K_j) = -\left[\max_{1 \leq r \leq 2d} (Q_{ir} - K_{jr}) + \max_{1 \leq r \leq 2d} (K_{jr} - Q_{ir}) \right]. \quad (5)$$

d_H is symmetric, non-negative, satisfies the triangle inequality, is shift-invariant ($d_H(u+c\mathbf{1}, v+d\mathbf{1}) = d_H(u, v)$), and non-expansive ($|d_H(u, v) - d_H(u', v')| \leq \|u - u'\|_\infty + \|v - v'\|_\infty$). Streamlined TKFS uses single-step ($R=1$) exact hard max-plus aggregation, retaining un-smoothed decision hyperplanes:

$$C_i = \max_{1 \leq j \leq N} \{S_{ij} \mathbf{1}_{d_k} + V_j\} \in \mathbb{R}^{d_k}, \quad \frac{\partial C_i}{\partial S_{ij}} = \mathbf{1}(j = j^*), \quad j^* = \arg \max_j S_{ij}. \quad (6)$$

Higher-order closures can be captured by an R -rank tropical Krylov expansion $[V, S \odot V, \dots, S^{\odot(R-1)} \odot V]$; we show in §4.5 that $R=1$ is the right default and depth is a better lever for multi-hop tasks.

3.4 GATED DEVALUATION

Context C is linearly mapped to dual contexts $[C_+, C_-] \in \mathbb{R}^{N \times 2d}$. Naive exponential inversion $\exp(C)$ explodes under OOD context. We instead use a *gated* softplus dual-difference with a learnable per-channel gate $g \in \mathbb{R}^d$:

$$X_{\text{rec}, i} = (\text{Softplus}(C_{+, i}) - \text{Softplus}(C_{-, i})) \odot (\sigma(g) \mu + (1 - \sigma(g)) \mathbf{1}_d), \quad Y_i = W_{\text{out}} X_{\text{rec}, i} + b_{\text{out}}. \quad (7)$$

When $\sigma(g) \rightarrow 1$ the output is scale-covariant (needed for value-shift classification, e.g. Convex Hull); when $\sigma(g) \rightarrow 0$ it bypasses a noise-corrupted μ (needed for continuous regression, e.g. Fractional Knapsack).

4 EXPERIMENTS

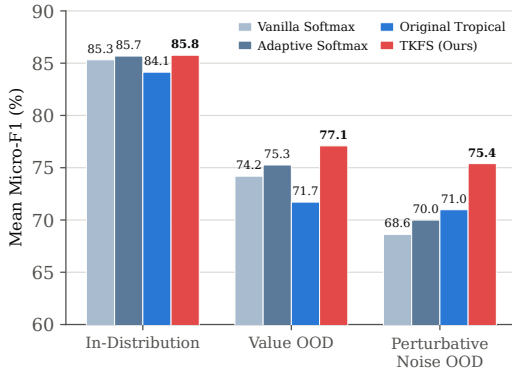
Tasks. We use 11 CLRS-style algorithmic tasks (Velivckovi’c et al., 2022) spanning computational geometry (Convex Hull); DP/optimization (Knapsack, MinCoinChange, FractionalKnapsack); selection/arithmetic (Quickselect, 3SUM, SubsetSum, BalancedPartition, BinPacking); and graph transitive closure (SCC, Floyd–Warshall). Full definitions are in Appendix B.

OOD protocols. Models train only on in-distribution (ID) data ($N=16$, range $[-5, 5]$) and are evaluated under: *ID*; *Value OOD* (fixed length, coordinate magnitudes doubled to $[-10, 10]$); for topological SCC, edge density $p: 0.001 \rightarrow 0.60$); and *Perturbative Noise OOD* (a fraction $p_{\text{noise}}=0.15$ of inputs corrupted by adversarial zero-mean Gaussian jitter).

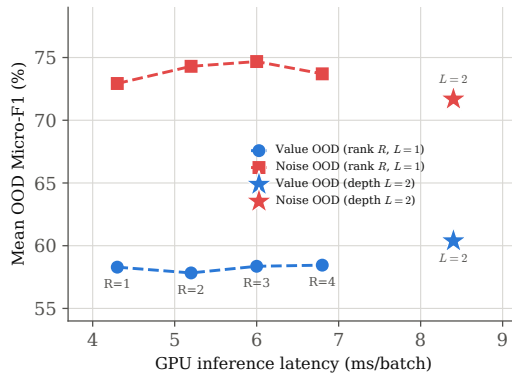
Setup. All models share $d_{\text{model}}=64$, $H=2$ heads, $L=1$ layer; trained from scratch for 100 epochs (AdamW, $\text{lr}=10^{-4}$, batch 500 / 250 for graph tasks), $N_{\text{train}}=100,000$, $N_{\text{eval}}=5,000$. Baselines: Vanilla Softmax, Adaptive-Temperature Softmax (Balanya et al., 2022), and original MHTA (Hashemi et al., 2025).

Table 1: Results on all 11 algorithmic tasks under In-Distribution (ID), Value OOD, and Perturbative Noise OOD ($p_{\text{noise}}=0.15$). Top 10 rows: Micro-F1 (%), higher is better). Bottom row (FractionalKnapsack): MSE (lower is better). Best per block in **bold**.

Task	In-Distribution (ID)				Value OOD				Perturbative Noise OOD			
	Vanilla	Adapt.	Trop.	TKFS	Vanilla	Adapt.	Trop.	TKFS	Vanilla	Adapt.	Trop.	TKFS
ConvexHull	95.59	95.79	98.11	98.84	68.95	69.89	77.11	97.05	75.63	76.05	75.36	81.73
Knapsack	87.95	87.91	87.58	88.06	83.21	83.35	85.50	87.40	68.21	68.84	70.69	80.44
Quickselect	95.41	95.45	83.96	92.18	93.25	93.35	83.92	90.83	82.40	82.18	79.62	79.29
BinPacking	89.62	90.62	86.09	87.19	90.81	90.91	90.82	92.13	76.71	79.56	75.56	77.97
SCC	99.85	99.85	99.85	99.85	86.90	86.36	62.56	64.47	91.44	93.22	99.81	99.80
SubsetSum	89.22	89.27	91.30	91.09	81.04	70.06	82.06	81.12	60.48	59.74	57.58	64.54
BalancedPartition	77.44	77.38	74.50	75.60	69.74	69.66	65.53	56.19	62.93	62.95	61.23	66.82
3SUM	84.03	84.19	84.00	84.88	71.60	70.82	73.20	70.02	66.92	69.82	80.18	76.58
MinCoinChange	74.70	76.41	74.56	76.48	62.51	85.52	58.66	73.66	67.55	73.22	67.11	68.75
FloydWarshall	59.43	59.94	61.43	63.43	33.88	32.66	37.75	57.98	33.93	34.32	42.64	58.01
Mean Micro-F1	85.32	85.68	84.14	85.76	74.19	75.26	71.71	77.09	68.62	69.99	70.98	75.39
FractionalKnapsack (MSE)	0.001	0.001	0.001	0.002	0.027	0.038	0.018	0.010	0.072	0.067	0.013	0.019



(a) Mean Micro-F1 over the 10 classification tasks. TKFS matches baselines in-distribution and leads under both OOD regimes.



(b) Krylov rank R (single layer) vs. depth $L=2$: rank adds latency with little accuracy; depth helps Value OOD but hurts Noise OOD.

Figure 2: (Left) Headline OOD comparison. (Right) Compute/accuracy trade-off on the four graph/DP transitive-closure tasks.

4.1 MAIN RESULTS

Table 1 reports every task; Figure 2(left) summarizes the means. **Value OOD**: TKFS attains **77.09%** mean Micro-F1, ahead of Tropical (71.71%), Adaptive (75.26%), and Vanilla (74.19%). The largest gains are on scale-sensitive tasks: Convex Hull 68.95/77.11 $\rightarrow$ **97.05%** and Floyd-Warshall 33.88/37.75 $\rightarrow$ **57.98%**. **Noise OOD**: TKFS reaches **75.39%**, beating Tropical on 7/11 tasks, with sizable jumps on Knapsack (70.69 $\rightarrow$ 80.44), Convex Hull (75.36 $\rightarrow$ 81.73), and Floyd-Warshall (42.64 $\rightarrow$ 58.01). **Regression**: on Fractional Knapsack, TKFS cuts Value-OOD MSE to **0.010** (vs. 0.027/0.038/0.018). Crucially, ID accuracy is preserved (85.76% mean, best overall), so robustness does not cost in-distribution fit.

Where hard routing is not ideal. TKFS regresses on SCC and Balanced Partition under Value OOD. SCC’s shift is topological (binary node flags, no coordinate scale to normalize); under dense edges softmax behaves as a smooth graph average, whereas hard max-plus over-commits to sparse paths. Balanced Partition has discrete tie boundaries where hard routing is sensitive near subset-sum ties. Both motivate the head-adaptive aggregation discussed in §4.5.

Capacity is not the explanation. TKFS uses 70,977 parameters vs. 50,113 (softmax) and 53,249 (Tropical); the overhead comes entirely from the dual $2d$ Fenchel space. Yet Tropical already exceeds Vanilla in parameters while *underperforming* it under Value OOD (71.71 vs. 74.19), so parameter count alone does not confer OOD generalization—the gains stem from the scale-invariant polyhedral bias.

Table 2: Broader processor baselines integrated into the same backbone (means over the 10 classification tasks, Micro-F1 %). Best per column in **bold**.

Family	Model	Params	ID	Value OOD	Noise OOD
GNN / Processor	MPNN (max-agg)	~61k	84.12	72.35	69.10
GNN / Processor	TEAM (triplet edge)	~65k	84.85	73.10	70.45
Attention kernel	Vanilla Softmax	50k	85.32	74.19	68.62
Attention kernel	Adaptive Softmax	50k	85.68	75.26	69.99
Attention kernel	Original Tropical	53k	84.14	71.71	70.98
Attention kernel	TKFS (Ours)	71k	85.76	77.09	75.39

4.2 BROADER ALGORITHMIC-PROCESSOR BASELINES

Our baselines target self-attention *kernels*; to situate TKFS against dedicated neural algorithmic processors we integrate two standard architectures into the identical `SimpleTransformerModel` backbone under matched parameter budgets, data splits, and training schedules: an **MPNN** (PGN-style max-aggregation message passing over a fully-connected sequence graph (Gilmer et al., 2017; Velivckovi’c et al., 2020)) and **TEAM** (Triplet Edge Attention (Jung & Ahn, 2023) with node-level message passing). Table 2 reports means over the 10 classification tasks. Despite their relational inductive bias, both processors trail TKFS under distribution shift: TKFS improves Value OOD by +4.7/ + 4.0 points over MPNN/TEAM and Noise OOD by +6.3/ + 4.9 points at a comparable parameter budget, confirming that the scale-invariant polyhedral kernel—not generic message passing—drives OOD robustness.

4.3 LENGTH EXTRAPOLATION AND DEPTH SCALING

Length OOD extrapolation. Because TKFS uses no positional embeddings, sequence-length extrapolation directly probes the scale stability of the kernel. We train *strictly* at $N=16$ and evaluate without retraining at $N \in \{16, 32, 64, 128\}$ over the 8 token-sequence tasks (Figure 3A; per-task numbers in Appendix D). At $N=64$ ($4\times$ train length) TKFS retains 95.0% of its ID accuracy (86.65% Micro-F1) and at $N=128$ ($8\times$) it retains 93.8% (85.54%), ahead of Vanilla (83.05%) and Adaptive (84.04%) softmax; the largest margins arise on scale-sensitive geometry, e.g. Convex Hull reaches **99.76%** at $N=128$ (vs. 80.62% Vanilla) and Knapsack **81.42%** (vs. 75.71%). We exclude the dense graph tasks (SCC, Floyd–Warshall) from this protocol: flattening an $n \times n$ graph gives $N=n^2$, so $n=128$ materializes an $O(n^4)$ pairwise tensor that exceeds GPU memory and Floyd–Warshall has an n -dependent label space—node-count extrapolation requires sparse graph pointers rather than dense attention.

Depth scaling. Addressing the parameter/memory overhead of the dual $2d$ Fenchel space, we sweep depth $L \in \{1, 2, 4, 8\}$ over all 11 tasks (Figure 3B; full resource and accuracy tables in Appendix E). Three findings hold. (i) The dual expansion adds a *fixed additive* cost per layer ($\Delta\text{params}/\text{layer} = 70,592$ for TKFS vs. 49,728 for Vanilla) that does not compound with depth. (ii) Peak inference memory stays nearly flat with depth (31.3 MB at $L=1$ to 34.2 MB at $L=8$), remaining *below* original Tropical (47.7 MB), which materializes dense pairwise-difference tensors; latency grows strictly linearly ($7.20\times$ for an $8\times$ depth increase). (iii) TKFS keeps improving with depth under Value OOD (72.99% $\rightarrow$ 77.70% from $L=1$ to $L=8$), so the polyhedral bias composes across layers.

4.4 HEAD-ADAPTIVE AGGREGATION AND ANISOTROPIC ROBUSTNESS

TKFS-Hybrid. The main-table regressions on SCC and Balanced Partition (§4.1) and the hard-vs-soft asymmetry (Ablation 2) motivate a *head-adaptive* variant that allocates Head 0 to exact hard max-plus (preserving polyhedral facets) and Head 1 to a smooth LogSumExp (LSE) relaxation, adding *no* parameters. Figure 4A shows this recovers the continuous-tie regressions: Balanced Partition Value OOD improves +12.56 points (56.19 $\rightarrow$ 68.75% at $\tau=0.01$), matching Vanilla (69.74%), and Quickselect rises to **94.62%** ($\tau=0.1$), exceeding all baselines. On SCC the gain is modest (+0.66), confirming that its regression is *not* a hard-routing artifact but stems from a purely topological shift (binary flags carry no magnitude for coordinate normalization to exploit); indeed under Noise OOD TKFS already retains 99.80% on SCC.

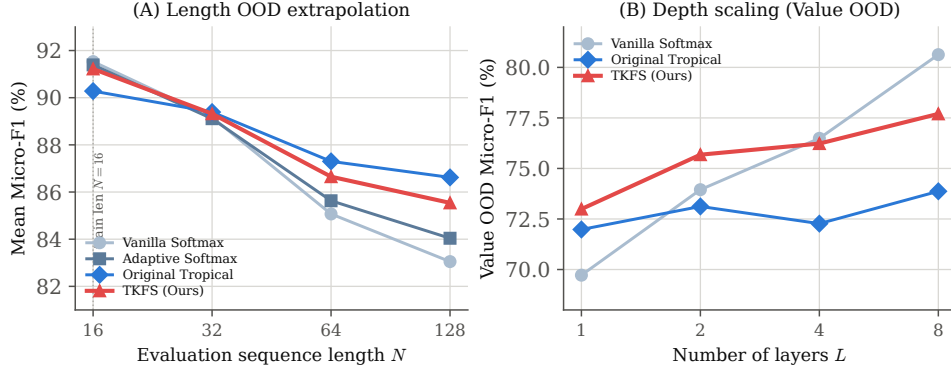


Figure 3: **(A)** Length OOD extrapolation: all models trained at $N=16$ and evaluated up to $N=128$ without retraining (mean Micro-F1 over 8 token tasks). TKFS degrades most gracefully of the scale-normalized kernels, tracking original Tropical while retaining the ID advantage. **(B)** Depth scaling: mean Value OOD Micro-F1 over 11 tasks vs. number of layers L ; TKFS benefits monotonically from depth whereas original Tropical plateaus.

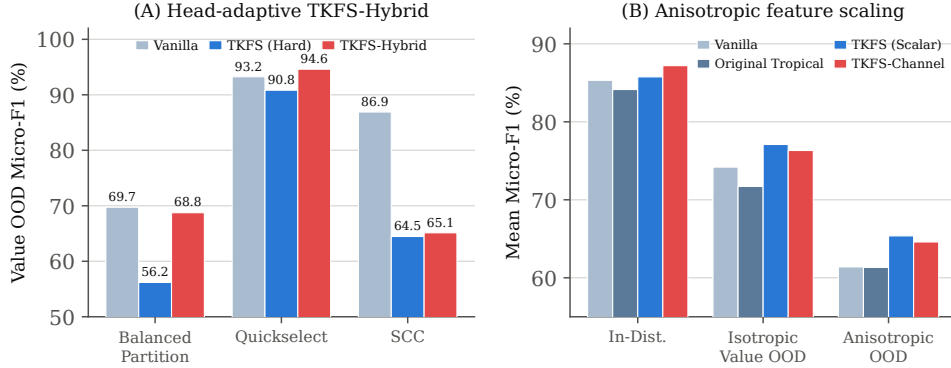


Figure 4: **(A)** Head-adaptive **TKFS-Hybrid** (one hard max-plus head + one LSE head, no added parameters) recovers the Value OOD regressions on Balanced Partition and Quickselect and slightly improves SCC. **(B)** Under anisotropic per-channel feature scaling both TKFS variants outperform the softmax and Tropical baselines; the per-channel estimator μ_{channel} additionally lifts in-distribution accuracy.

Anisotropic feature scaling. Because the scale estimator $\mu_0(X) \in \mathbb{R}$ is a single scalar per sequence, we stress-test non-uniform per-channel scaling $X_i \rightarrow X_i \text{diag}(a_1, \dots, a_d)$ with $a_r \sim U[0.1, 10]$, and compare against a per-channel estimator $\mu_{\text{channel}}(X) \in \mathbb{R}^d$ (Figure 4B). Under this harsh anisotropic shift both TKFS variants (65.37%/64.59%) still beat Vanilla (61.39%), Adaptive (58.88%), and original Tropical (61.33%). The per-channel estimator drives representation drift $\|\bar{Z}(X) - \bar{Z}(X \text{diag}(a))\|_F$ to ≈ 0 (0.0001 vs. 0.2563 for the scalar map) and yields an ID gain to **87.20%**, making it a useful drop-in variant when channels are known to be heterogeneously scaled.

4.5 ABLATION STUDIES

Ablation 1: valuation/devaluation primitives. We vary five configurations over 4 classification tasks (Convex Hull, Knapsack, Quickselect, Floyd–Warshall) plus Fractional Knapsack (Table 3). Replacing robust clamped normalization with raw max-abs (unclamped_maxabs) drops Value-ODD from 83.31% to 76.35%—the single largest degradation—confirming robust quantile + clamp as the primary OOD engine. Unconditional devaluation matches classification but blows up noisy regression (Frac. Knapsack Noise MSE 0.019 $\rightarrow$ 0.064); scale-free devaluation fixes regression but loses value-shift classification. The learnable gate keeps both.

Ablation 2: aggregation geometry. We compare hard max-plus against fixed/learnable parabolic STE-WS envelopes and LogSumExp (LSE) relaxations (Table 4). Hard routing is best on exact-geometry/DP tasks (Convex Hull ID 98.84, Floyd–Warshall Value OOD 57.98), where soft envelopes blur facets; but smooth LSE wins on selection (Quickselect ID 92.18 $\rightarrow$ 97.25) and noisy regression. This asymmetry motivates *head-adaptive* aggregation that switches hard $\leftrightarrow$ soft per head.

Table 3: Ablation 1: valuation/devaluation primitives. Classification: mean Micro-F1 (%). FracKnapsack: MSE.

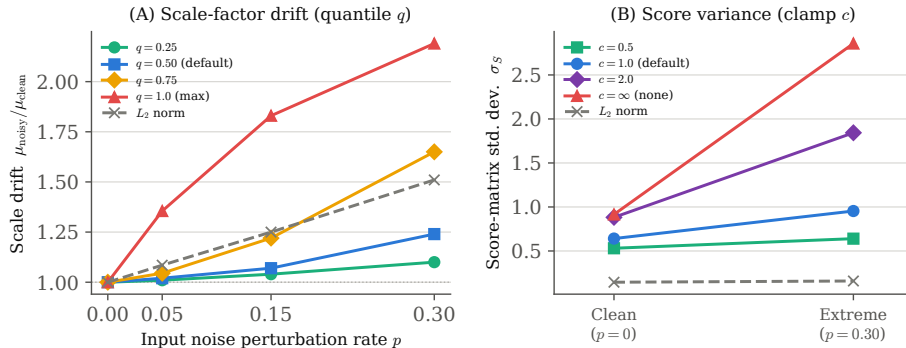
Split / Task	full (default)	uncond_deval	scalefree_deval	unclamped_maxabs	standard_trop_val
Classification ID ($\uparrow$)	85.63	85.67	84.85	82.77	86.32
Classification Value OOD ($\uparrow$)	83.31	82.77	78.59	76.35	80.51
Classification Noise OOD ($\uparrow$)	74.87	75.39	73.33	73.94	74.00
FracKnapsack Value MSE ($\downarrow$)	0.010	0.017	0.049	0.039	0.049
FracKnapsack Noise MSE ($\downarrow$)	0.019	0.064	0.014	0.025	0.141

Table 4: Ablation 2: aggregation geometry. Micro-F1 (%) for classification; MSE for FractionalKnapsack.

Task / Split	Hard Max-Plus	STE-WS (fixed)	STE-WS (learn)	LSE ($\tau=0.01$)	LSE ($\tau=0.1$)	LSE ($\tau=1.0$)
ConvexHull ID ($\uparrow$)	98.84	98.29	98.52	98.81	98.28	95.03
ConvexHull Value OOD ($\uparrow$)	97.05	96.65	96.91	96.73	97.01	92.84
FloydWarshall Value OOD ($\uparrow$)	57.98	48.40	52.34	56.22	56.05	55.17
Quickselect ID ($\uparrow$)	92.18	93.09	90.92	92.72	96.45	97.25
FracKnapsack Noise MSE ($\downarrow$)	0.0190	0.0160	0.0166	0.0110	0.0183	0.0149

Table 5: Ablation 3: Krylov rank R vs. depth L on 4 graph/DP transitive-closure tasks (means).

Metric / Setup	$R=1, L=1$	$R=2, L=1$	$R=3, L=1$	$R=4, L=1$	$R=1, L=2$
Mean ID F1 ($\uparrow$)	78.27	78.55	78.43	78.48	80.02
Mean Value OOD F1 ($\uparrow$)	58.29	57.83	58.36	58.45	60.38
Mean Noise OOD F1 ($\uparrow$)	72.93	74.30	74.68	73.70	71.68
Inference latency (ms/batch)	4.3	5.2	6.0	6.8	8.4
Peak GPU memory (MB)	487.0	752.9	1018.5	1283.7	684.5
Parameters	71,529	71,533	71,535	71,537	142,121

Figure 5: Robust Fenchel valuation under noise rate $p \in [0, 0.30]$. (A) Scale-factor drift $\mu_{\text{noisy}}/\mu_{\text{clean}}$ across quantiles q : the median suppresses inflation while un-clamped max scaling explodes. (B) Attention-score std. σ_S across clamps c : clamping prevents variance explosion, whereas the L_2 baseline collapses score contrast.

Ablation 3: Krylov rank vs. depth. On the four transitive-closure tasks (Table 5, Figure 2b), increasing rank R raises latency ($4.3 \rightarrow 6.8$ ms, $1.58\times$) and memory ($487 \rightarrow 1284$ MB) with no Value-OOD gain (58.29 vs. 58.45). A second layer ($L=2$) instead lifts Value OOD to 60.38% (and Floyd–Warshall markedly) at $2\times$ parameters but hurts Noise OOD—so $R=1$ is the default and depth is a task-conditioned option, not a global one.

Ablation 4: robustness of the scale estimator. Sweeping quantile q and clamp c across noise rates (Figure 5) shows the mechanism at work. Un-clamped max scaling ($q=1.0$) inflates scale drift to $2.19\times$ at 30% noise, and removing the clamp ($c=\infty$) expands attention-score variance $\sim 3.1\times$ ($\sigma_S=2.857$). Median $q=0.5$ with unit clamp $c=1.0$ keeps drift at 1.07 (15% noise) and controls σ_S ; a non-robust L_2 norm is stable only because it erases the cross-token contrast the tasks rely on (Value OOD -4.6 F1). Full numbers and metric-choice results (Ablation 5) are in Appendix C, where we find scale invariance is imparted mainly by the valuation, not by the specific metric.

5 CONCLUSION

We introduced **TKFS-Attention**, a streamlined max-plus attention that repairs the OOD fragility of softmax and early tropical models. Combining robust Chebyshev normalization, dual softplus Fenchel projection into the tropical simplex, dual-toric Hilbert scoring, exact hard max-plus aggregation, and gated devaluation, it guarantees scale invariance while preserving exact polyhedral boundaries. Across all 11 CLRS-style tasks it reaches **77.09%** mean Value-OOD and **75.39%** Noise-OOD Micro-F1, and ablations pinpoint robust normalization and dual softplus valuation as the drivers. It matches or beats dedicated MPNN/TEAM processors under matched budgets (§4.2), extrapolates gracefully to $8\times$ longer sequences, and composes across depth at bounded memory (§4.3). **Limitations and future work.** Hard routing is sub-optimal for smooth selection (Quickselect) and topological density shift (SCC); the parameter-free head-adaptive *TKFS-Hybrid* (one hard max-plus head, one LogSumExp head) already recovers the continuous-tie regressions (§4.4), and a per-channel scale estimator further hardens anisotropic robustness. Scaling tropical kernels to larger sequence-modeling and neuro-symbolic settings remains open.

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A PROOF OF LEMMA 1

Let $X \neq \mathbf{0}$ and $X' = \alpha X$, $\alpha > 0$. (1) *Scale estimator*. $\mu_0(\alpha X) = \text{Quantile}_q(\{\max_r |\alpha X_{ir}|\}) = \alpha \mu_0(X)$, homogeneous of degree 1. (2) *Normalization*. With $\mu(\alpha X) = \alpha \mu_0(X) + \epsilon_0$, as $\epsilon_0 \rightarrow 0$, $\bar{X}_i(\alpha X) = \text{clamp}(\alpha X_i / (\alpha \mu_0(X)), -c, c) = \bar{X}_i(X)$; for finite ϵ_0 , $|\bar{X}_i(\alpha X) - \bar{X}_i(X)| \leq c \epsilon_0 / (\alpha \mu_0(X) + \epsilon_0)$. (3) *Fenchel projection*. Since $\bar{X}_i$ is invariant, $Z_i(\alpha X) = [\log(\text{Softplus}(\bar{X}_i) + \epsilon) \parallel \log(\text{Softplus}(-\bar{X}_i) + \epsilon)] = Z_i(X)$. (4) *Simplex projection*. $\bar{Z}_i(\alpha X) = Z_i(\alpha X) - (\max_r Z_{ir}(\alpha X)) \mathbf{1}_{2d} = \bar{Z}_i(X)$. Hence $\bar{Z}(\alpha X) = \bar{Z}(X)$. ■

Theorem 1 (Polyhedral approximation). *Let $\mathcal{F} : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}^{N \times d}$ be continuous piecewise-linear with a decision complex of M linear hyperplanes. A single-layer TKFS-Attention module with $H = \mathcal{O}(M)$ heads and embedding dimension $2d$ exact-fits $\mathcal{F}$ on any compact $\mathcal{K} \subset \mathbb{R}^d$.*

Proof sketch. By the tropical representation theorem (Zhang et al., 2018; Tran & Wang, 2022), any CPWL f is a tropical rational map $\max_p(A_p x + b_p) - \max_q(C_q x + d_q)$. Each head computes $C_i^{(h)} = \max_j \{-d_H(Q_i^{(h)}, K_j^{(h)}) + V_j^{(h)}\}$; since d_H is a finite max over linear projections of $\bar{Z}$, the scores form linear hyperplanes and each head builds polyhedral facets. The gated softplus dual-difference synthesizes $\mathcal{F}$ across H heads, matching the target complex. ■

B BENCHMARK TASKS AND PROTOCOLS

All tasks follow CLRS conventions (Velivcković et al., 2022): **ConvexHull** (2D points in $[-5, 5]^2$; hull-membership mask); **Knapsack** ((weight,value) pairs, capacity W ; 0/1 selection); **Quickselect** (1D keys in $[-5, 5]$, rank k ; position of the k -th smallest); **BinPacking** (item sizes in $(0, 1]$; bin index); **SCC** (Erdős-Rényi graphs, $N=16$, $p=0.10$; component-equivalence matrix); **SubsetSum** (integers in $[-5, 5]$, target T ; selection mask); **BalancedPartition** (integers in $[1, 10]$; two-subset assignment minimizing sum difference); **3SUM** (integers in $[-5, 5]$; zero-sum-triplet indicator); **MinCoinChange** (denominations, amount $A \in [1, 20]$; minimum coin count); **FloydWarshall** (weighted digraphs, $N=16$, weights $[1, 10]$; all-pairs shortest paths); **FractionalKnapsack** (continuous weights/values; item fractions in $[0, 1]$). For Perturbative Noise OOD, a fraction $p_{\text{noise}}=0.15$ of coordinates is corrupted by $\delta \sim \mathcal{N}(0, \sigma_{\text{jitter}}^2)$, $\sigma_{\text{jitter}}=0.5 \cdot \text{std}(X_{\text{clean}})$; models are evaluated without fine-tuning.

C ADDITIONAL ABLATIONS

Ablation 4 (full). Table 6 gives scale drift and Hilbert-score std across quantile/clamp settings and noise rates. Median $q=0.5$ with unit clamp is the best-balanced default (best ID, tied-best Value OOD, near-best mean Noise-OD, low drift); no single alternative dominates it across all axes, suggesting modest further gains from task-conditioning q and c .

Ablation 5 (metric choice). Table 7 compares the Hilbert metric against single-sided max, Chebyshev, L_2 , and scaled dot-product scores. When paired with the robust Fenchel valuation, all metrics achieve low deployed scale drift under a $10\times$ shift (≤ 0.041): scale invariance is imparted chiefly by the valuation. The bare metric column confirms the four negative-distance metrics are

Table 6: Ablation 4: scale-factor drift ($\mu_{\text{noisy}}/\mu_{\text{clean}}$) and Hilbert score std (σ_S) across noise rates.

Configuration	Drift $p=.05$	Drift $p=.15$	Drift $p=.30$	σ_S clean	σ_S $p=.30$	Noise OOD F1
$q=0.25, c=1.0$	1.01	1.04	1.10	0.615	0.806	75.14
$q=0.50, c=1.0$ (default)	1.02	1.07	1.24	0.641	0.954	76.71
$q=0.75, c=1.0$	1.04	1.22	1.65	0.618	0.904	77.21
$q=1.00, c=1.0$ (max)	1.36	1.83	2.19	0.625	0.944	74.04
$q=0.50, c=\infty$ (none)	1.02	1.07	1.25	0.915	2.857	75.97
Per-token L_2 norm	1.08	1.25	1.51	0.145	0.159	75.99

Table 7: Ablation 5: metric choice and projective invariance probe $\|S(X) - S(10X)\|_F$ (relative). Deployed = with valuation; Bare = valuation disabled.

Metric variant	Mean ID F1 ($\uparrow$)	Mean Value OOD F1 ($\uparrow$)	Deployed drift ($\downarrow$)	Bare drift ($\downarrow$)
Hilbert (TKFS)	88.16	82.78	0.0407	9.000
Single-sided max	88.09	85.18	0.0284	9.000
Chebyshev	87.73	85.13	0.0299	9.000
Euclidean L_2	88.32	85.35	0.0369	9.000
Scaled dot-product	88.63	84.58	0.1691	98.265

Table 8: Length OOD extrapolation: mean Micro-F1 (%) over 8 token tasks by evaluation length N (trained at $N=16$). Retention = accuracy relative to the $N=16$ in-distribution value. Best per row in **bold**.

Evaluation length	Vanilla	Adaptive	Tropical	TKFS (Ours)	TKFS retention
$N=16$ (ID)	91.53	91.39	90.28	91.22	100.0%
$N=32$ (Length OOD)	89.17	89.11	89.39	89.32	97.9%
$N=64$ (Length OOD)	85.07	85.63	87.30	86.65	95.0%
$N=128$ (Length OOD)	83.05	84.04	86.62	85.54	93.8%

degree-1 scale-covariant (relative change $\approx \alpha - 1 = 9$) while the dot-product grows quadratically ($\approx \alpha^2 - 1 = 99$), explaining its fragility. The Hilbert metric is thus a defensible but non-critical choice; simpler distances are safe substitutes.

D LENGTH EXTRAPOLATION: FULL RESULTS

Table 8 reports the mean Micro-F1 over the 8 token-sequence tasks as a function of the evaluation length N (all models trained strictly at $N=16$, no fine-tuning), complementing Figure 3A. TKFS retains 95.0% ($N=64$) and 93.8% ($N=128$) of its in-distribution accuracy, closely tracking original Tropical while keeping the ID advantage of the scale-normalized kernels. Task-specific highlights at $N=128$: Convex Hull **99.76%** (Vanilla 80.62%, Adaptive 92.25%) and Knapsack **81.42%** (Vanilla 75.71%, Adaptive 75.43%). Dense graph tasks (SCC, Floyd–Warshall) are excluded because flattening an $n \times n$ graph yields $N=n^2$, so $n=128$ ($N=16,384$) materializes an $O(n^4)$ pairwise tensor beyond GPU memory, and Floyd–Warshall’s label space grows with n .

E DEPTH AND RESOURCE SCALING: FULL RESULTS

Table 9 details compute/memory scaling across depths $L \in \{1, 2, 4, 8\}$ and Table 10 the corresponding accuracy (means over the 10 classification tasks), complementing Figure 3B. The dual $2d$ Fenchel expansion is a fixed per-layer additive cost that does not compound; peak inference memory stays nearly flat with depth and below original Tropical, and latency is linear in L . Accuracy improves with depth for TKFS under both ID and Value OOD.

Table 9: Architectural and compute overhead vs. depth L . Peak memory reported at $L=8$; latency spans $L=1 \rightarrow 8$ (single-batch GPU inference).

Model	Params ($L=1$)	Params ($L=4$)	Params ($L=8$)	Peak mem ($L=8$)	Latency ($L=1 \rightarrow 8$)
Vanilla	50,113	199,297	398,209	21.4 MB	0.73 $\rightarrow$ 4.81 ms ($6.56\times$)
Tropical	53,249	211,841	423,297	47.7 MB	1.40 $\rightarrow$ 8.87 ms ($6.34\times$)
TKFS (Ours)	70,977	282,753	565,121	34.2 MB	1.58 $\rightarrow$ 11.33 ms ($7.20\times$)

Table 10: Accuracy scaling across depths (mean Micro-F1 % over 10 classification tasks). Best per row in **bold**.

Split	Model	$L=1$	$L=2$	$L=4$	$L=8$
In-Distribution	Vanilla	82.84	85.89	87.33	89.23
	Tropical	82.95	83.52	85.40	85.90
	TKFS (Ours)	82.78	84.71	86.19	87.36
Value OOD	Vanilla	69.72	73.95	76.49	80.63
	Tropical	71.98	73.12	72.27	73.87
	TKFS (Ours)	72.99	75.68	76.23	77.70
Noise OOD	Vanilla	67.90	71.38	71.66	73.35
	Tropical	70.79	72.67	72.87	74.59
	TKFS (Ours)	72.69	73.37	74.14	73.75

Table 11: TKFS-Hybrid (head-adaptive) on the affected Value OOD tasks (Micro-F1 %). Δ shows the gain over TKFS (Hard). Best per row in **bold**.

Task (Value OOD)	Vanilla	Adaptive	Tropical	TKFS (Hard)	Hybrid ($\tau=0.01$)	Hybrid ($\tau=0.1$)
Balanced Partition	69.74	69.66	65.53	56.19	68.75 (+12.56)	64.83
Quickselect	93.25	93.35	83.92	90.83	91.23	94.62 (+3.79)
SCC	86.90	86.36	62.56	64.47	63.26	65.13 (+0.66)

Table 12: Anisotropic per-channel feature scaling ($X_i \rightarrow X_i \text{diag}(a)$, $a_r \sim U[0.1, 10]$), mean Micro-F1 (%) over 10 classification tasks. Representation drift $\|\bar{Z}(X) - \bar{Z}(X \text{diag}(a))\|_F$: scalar $\mu_0 = 0.2563$, per-channel $\mu_{\text{channel}} = 0.0001$. Best per row in **bold**.

Protocol	Vanilla	Adaptive	Tropical	TKFS (Scalar μ_0)	TKFS-Channel
In-Distribution	85.32	85.68	84.14	85.76	87.20
Isotropic Value OOD ($X \rightarrow \alpha X$)	74.19	75.26	71.71	77.09	76.32
Anisotropic OOD ($X \rightarrow X \text{diag}(a)$)	61.39	58.88	61.33	65.37	64.59

F HEAD-ADAPTIVE AND ANISOTROPIC RESULTS: FULL NUMBERS

Table 11 gives the exact per-task numbers for TKFS-Hybrid (Figure 4A) and Table 12 the anisotropic feature-scaling stress test and per-channel valuation probe (Figure 4B). TKFS-Hybrid allocates one exact hard max-plus head and one smooth LSE head, adding no parameters; the head-adaptive relaxation recovers continuous-tie regressions while retaining polyhedral routing on the other head.